

High-LevelDesign

Legal Document Customer Support RAG

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Internship Project High-Level Design (HLD) Report

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1 System Overview

Problem Definition

Users often struggle to extract specific information from large documents such as reports or manuals. Reading entire documents is time-consuming and inefficient.

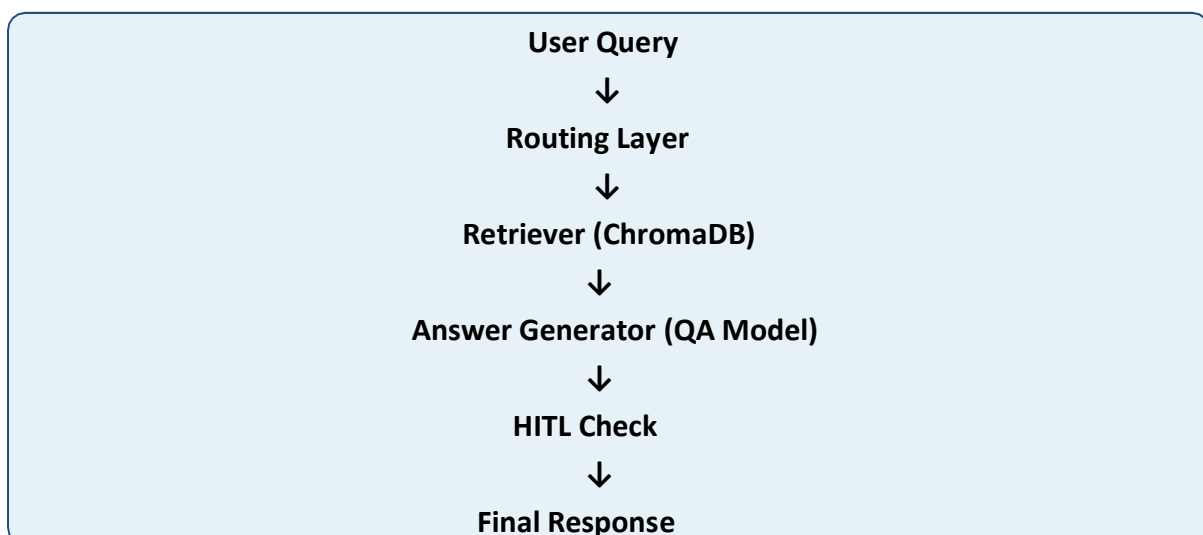
Objective

1. Process PDF documents
2. Retrieve relevant information using embeddings
3. Generate accurate answers based on context
4. Provide fallback using Human-in-the-Loop (HITL)

Scope

- **Input:** Legal PDF documents
- **Output:** Contextual answers to user queries
- **Interface:** CLI-based system
- Includes Human-in-the-Loop(HITL)escalation

2 Architecture Diagram



3 Component Description

Document Loader

Loads PDF documents using Py PDF Loader.

Chunking Strategy

- Recursive Character Text Splitter
- Chunksize:500
- Overlap:100

Embedding Model

- sentence-transformers(all-MiniLM-L6-v2)
- Converts text into vector embeddings

Vector Store

ChromaDB is used to store embeddings locally for fast retrieval

Retriever

Fetches top relevant chunks using similarity search.

LLM

- llama-3.3-70b-versatile(GroqAPI)
- Generates answers using retrieved context

Graph Work flow Engine

Built using Lang Graph to control work flow execution.

Routing Layer

- Decides response path
- Routes to HITL if needed

HITL Module

Handles escalation and returns human-generated responses.

4 Data Flow

Document Processing Flow

PDF→Load→Chunk→Embed→Store

QueryFlow

User Query → Routing → Retrieval → Answer Generation → Response

5 Technology Choices

Chroma DB

- Light weight
- Easy integration
- Local storage

LangGraph

- Workflow-based execution
- Conditional routing

Groq

- Fast inference
- High-quality reasoning

6 Scalability Considerations

Handling Large Documents

Chunking ensures efficient processing of large PDFs.

Increasing Query Load

System can be scaled using better models and optimized retrieval.

Latency Optimization

- Limit top-k retrieval
- Use efficient embedding models
- Reduce unnecessary context